Advanced Mechanics

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1 Lagrangian Mechanics

1.1 Principle of Least Action and the Euler-Lagrange Equations

When all the forces are derivable from a potential (conservative forces), a particle picks a trajectory which extremizes the action functional given by,

$$S = \int_t \mathcal{L}(q, \dot{q}, t) dt, \tag{1}$$

where the Lagrangian function is given as,

$$\mathcal{L} = T - V. \tag{2}$$

The critical point of (1) is a trajectory $q(t)$, and finding it is a problem in the calculus of variations. The general result which is applicable to many situations is the Euler-Lagrange equation given as,

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \frac{\partial \mathcal{L}}{\partial q} = 0. \tag{3}$$

It should be noted that if the system has n generalized coordinates q_n, then there are n Euler-Lagrange equations for each generalized coordinate. The routine to derive the Euler-Lagrange equations is summarized below,

$$\delta S = \int_t \left(\frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial \dot{q}}\delta \dot{q}\right) dt, \tag{4}$$

$$\delta S = \int_t \left(\frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial \dot{q}}(\delta q)_t\right) dt, \tag{5}$$

$$\delta S = \frac{\partial L}{\partial \dot{q}}\delta q|_{t_i}^{t_f} + \int_t \left(\frac{\partial L}{\partial q}\delta q - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right)\delta q\right) dt. \tag{6}$$

The term outside the integral disappears since $\delta q = 0$ vanishes at the boundary points. The integrand must vanish for all variations δq. In general this can be showed using the method of contradiction, taking δq to be some compact function, that the integrand in such cases must always vanish identically. The integrand is nothing but the Euler-Lagrange equation.

1.2 Conserved Quantities

1.2.1 Generalized Momentum

We define the generalized momentum as,

$$p = \frac{\partial \mathcal{L}}{\partial \dot{q}}, \tag{7}$$

so with this definition (3) turns into,

$$\dot{p} = \frac{\partial \mathcal{L}}{\partial q}, \tag{8}$$

and can be thought of as a generalization of $F = \dot{p}$. Thus if the generalized force is zero, i.e, the Lagrangian does not depend on one of the coordinates, then the corresponding momentum is conserved. Such a coordinate is called **cyclic or ignorable**.

1.2.2 Energy Function

Energy also plays a crucial part in describing mechanical systems. To construct a definition for energy in terms of generalized coordinates, consider the time derivative of the Lagrangian function,

$$\frac{dL}{dt} = \frac{\partial L}{\partial q}\dot{q} + \frac{\partial L}{\partial \dot{q}}\ddot{q} + \frac{\partial L}{\partial t} = p\dot{q} + \dot{p}\ddot{q} + \frac{\partial L}{\partial t}, \tag{9}$$

$$\frac{d}{dt}\left(p\dot{q} - L\right) = -\frac{\partial L}{\partial t}, \tag{10}$$

which prompts the definition of the energy function,

$$h(q, \dot{q}, t) = p\dot{q} - L. \tag{11}$$

The energy function remains conserved as long as the Lagrangian does not depend explicitly on time. As it will turn out, the energy function is the Hamiltonian of the system but expressed in terms of the generalized velocity, where as the Hamiltonian function trades it for generalized momentum through the Legendre transformation.

1.3 Case Study: The Kepler Problem

As an example of the power of the Lagrangian method to solve mechanical problems, we will explore the two-body problem in Newtonian gravity. First of all we will reduce (pun-intended!) the system to a single body of reduced mass μ rotating around a bigger body with total mass M. The centre of mass is placed at the origin and the relative displacement vector is,

$$\vec{r} = \vec{r_2} - \vec{r_1}, \tag{12}$$

with corresponding relative velocity vector,

$$\vec{v} = \vec{v_2} - \vec{v_1}, \tag{13}$$

which has the squared speed,

$$v^2 = v_2^2 + v_1^2 - 2\vec{v_1} \cdot \vec{v_2} = v_2^2 + v_1^2 + 2v_1 v_2. \tag{14}$$

I have used that the individual speeds are always directed opposite to each other. This is in fact comes from the conservation of linear momentum of the entire two body system, since there is no external force. Conservation of linear momentum also gives us,

$$m_1\vec{v_1} + m_2\vec{v_2} = 0 \implies \frac{m_1}{m_2} = \frac{v_2}{v_1}. \tag{15}$$

A corresponding similar expression for the orbital separations is,

$$\frac{m_1}{m_2} = \frac{r_2}{r_1}. \tag{16}$$

Now before we write down the Lagrangian of the two-body system, we will do some intermediate work to express v_1^2 and v_2^2 as functions of v alone. Proceed as follows for the former,

$$v^2 = v_1^2\frac{m_1^2}{m_2^2} + v_1^2 + 2v_1^2\frac{m_1}{m_2} = v_1^2\frac{M^2}{m_2^2} \implies v_1^2 = \frac{v^2 m_2^2}{M^2}, \tag{17}$$

with the similar expression for the latter,

$$v_2^2 = \frac{v^2 m_1^2}{M}. \tag{18}$$

Plugging these latest results into the general Lagrangian, and that the reduced mass μ is defined as,

$$\mu = \frac{m_1 m_2}{M}, \tag{19}$$

$$\mathcal{L} = \frac{1}{2}\mu v^2 + \frac{GM\mu}{r}. \tag{20}$$

Now we shall introduce the two generalized coordinates,

$$\mathcal{L} = \frac{1}{2}\mu \dot{r}^2 + \frac{1}{2}\mu r^2 \dot{\phi}^2 + \frac{GM\mu}{r}, \tag{21}$$

and the corresponding equations of motion,

$$r\dot{\phi}^2 - \frac{GM}{r^2} = \ddot{r}, \quad \mu r^2 \dot{\phi} = L, \tag{22}$$

where L is the conserved angular momentum for the reduced mass. To solve the equations of motion however, we will appeal to the conservation of energy since the Lagrangian does not depend on time. The energy function is given as,

$$E = \sum_{r,\phi} p\dot{q} - \mathcal{L} = \frac{1}{2}\mu \dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{GM\mu}{r}, \tag{23}$$

and here we can define a specific energy and specific angular momentum by dividing the above equation by the reduced mass. Doing so we get,

$$2\epsilon = \dot{r}^2 + 2\left[\frac{h^2}{2r^2} - \frac{K}{r}\right], \quad K = GM, \quad \epsilon = \frac{E}{\mu}, \quad h = \frac{L}{\mu}. \tag{24}$$

Now to convert this into an inhomogeneous oscillator problem, we can use the substitution,

$$u = \frac{1}{r}, \tag{25}$$

and we start by taking the differential of u as

$$\frac{du}{d\phi} = \frac{du}{dr}\frac{dr}{d\phi} = \frac{-1}{r^2}\frac{dr}{d\phi} \implies \frac{dr}{d\phi} = -r^2\frac{du}{d\phi}. \tag{26}$$

$$\dot{r} = \frac{dr}{dt} = \frac{dr}{d\phi}\frac{d\phi}{dt} = -r^2\frac{du}{d\phi}\frac{d\phi}{dt} \tag{27}$$

Substituting that $\dot{\phi} = \frac{h}{r^2}$ and squaring we get,

$$\dot{r}^2 = h^2 \left(\frac{du}{d\phi}\right)^2, \tag{28}$$

and plugging this result into the equation for specific energy we have,

$$h^2 \left(\frac{du}{d\phi}\right)^2 + 2\left(-\frac{K}{r} + \frac{h^2}{2r^2}\right) = 2\epsilon, \tag{29}$$

which after trading r for u gives,

$$\left(\frac{du}{d\phi}\right)^2 - \frac{2Ku}{h^2} + u^2 = \frac{2\epsilon}{h^2}, \tag{30}$$

and taking a derivative we get,

$$2\left(\frac{du}{d\phi}\right)\frac{d^2u}{d\phi^2} - \frac{2K}{h^2}\frac{du}{d\phi} + 2u\frac{du}{d\phi} = 0. \tag{31}$$

Cleaning everything up we get that,

$$\frac{d^2u}{d\phi^2} - \frac{K}{h^2} + u = 0, \tag{32}$$

$$\frac{d^2u}{d\phi^2} + u = \frac{K}{h^2}. \tag{33}$$

This is an oscillatory problem with the general solution,

$$u = \frac{1 + ecos(\phi - \phi_0)}{p} \implies r = \frac{p}{1 + ecos(\phi - \phi_0)}, \tag{34}$$

with the constants equal to,

$$p = \frac{h^2}{K}, \quad e = \sqrt{1 + \frac{2\epsilon h^2}{K^2}}. \tag{35}$$

The general solution is the general equation for a conic section in polar coordinates. For $0 < e < 1$ we get bound orbits: 1 circle and a family of ellipses. These bound orbits follow Kepler's Third Law,

$$P^2 = \frac{4\pi^2 a^3}{GM}, \tag{36}$$

which follows from the second law,

$$\frac{dA}{dt} = \frac{L}{2\mu}, \tag{37}$$

with energy and angular momentum given by,

$$E = -\frac{GM\mu}{2a}, \quad L = \mu\sqrt{GMa(1-e^2)}, \tag{38}$$

and the general solution,

$$r = \frac{a(1-e^2)}{1+ecos\theta}. \tag{39}$$

The minimum energy of escape orbit is the parabolic orbit. These have vanishing energy (and e) and the general solution,

$$r = \frac{2a}{1+ecos\theta}, \tag{40}$$

where as hyperbolic orbits have $e > 1$ and the general solution,

$$r = \frac{a(e^2-1)}{1+ecos\theta}. \tag{41}$$

1.4 Invariance of the Euler-Lagrange Equations

If we modify the Lagrangian function by the amount,

$$\mathcal{L}^{'} = \mathcal{L} + \Delta\mathcal{L}, \quad \Delta\mathcal{L} = \frac{df(q,t)}{dt}, \tag{42}$$

then the equations of motion remain invariant. We can do this by a brute force computation or by noting that the new action will be

$$S^{'} = S + \int_{t_1}^{t_2} \Delta\mathcal{L}dt, \tag{43}$$

$$S^{'} = S + \int_{t_1}^{t_2} \frac{df}{dt}dt = S + f(q(t2),t2) - f(q(t1),t1). \tag{44}$$

The extremization will then read as,

$$\delta S^{'} = \delta S + \delta f(q(t2),t2) - \delta f(q(t1),t1), \tag{45}$$

but the assumption is that the endpoints of variation remain fixed, thus we have

$$\delta S^{'} = \delta S, \tag{46}$$

and the Euler-Lagrange equations must then be the same. Below is the brute force approach.

$$\frac{d}{dt}\left(\frac{\partial L^{'}}{\partial \dot{q}}\right) - \frac{\partial L^{'}}{\partial q} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} + \frac{d}{dt}\left(\frac{\partial \Delta L}{\partial \dot{q}}\right) - \frac{\partial \Delta L}{\partial q}, \tag{47}$$

and if we want the left hand side to be zero, we want that

$$\frac{d}{dt}\left(\frac{\partial \Delta L}{\partial \dot{q}}\right) - \frac{\partial \Delta L}{\partial q} = 0, \tag{48}$$

Start by writing out,

$$\Delta L = \frac{df}{dt} = \frac{\partial f}{\partial q}\dot{q} + \frac{\partial f}{\partial t}, \tag{49}$$

and it is trivial then to show that the Euler-Lagrange equation for the change in the Lagrangina is zero. It also follows that the momenta and energy do not remain the same though and are expressed as,

$$p^{'} = p + \frac{\partial f}{\partial q}, \quad h^{'} = h - \frac{\partial f}{\partial t}. \tag{50}$$

1.5 Motion in a Rotating Reference Frame

1.5.1 Motion on a Turntable

Consider a coordinate system S that rotates with constant angular velocity Ω. At $t = 0$, its x and y axes coincide with an inertial, static reference frame $S^{'}$. Both S and $S^{'}$ have coincident z axes. The coordinates therefore transform as,

$$x^{'} = xcos(\Omega t) - ysin(\Omega t), \quad y^{'} = ycos(\Omega t) + xsin(\Omega t), \quad z^{'} = z, \tag{51}$$

and the squared speed is obtained as,

$$(v^{'})^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2 - 2\Omega(y\dot{x} - x\dot{y}) + \Omega^2(x^2 + y^2), \tag{52}$$

which ultimately gives the Lagrangian,

$$L = \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2 + m\dot{z}^2\right) - \Omega(y\dot{x} - x\dot{y}) + \frac{1}{2}m\Omega^2(x^2 + y^2) - V(x, y, z). \tag{53}$$

Dumping this Lagrangian into the Euler-Lagrange equations gives the set of vectorized equations,

$$m\ddot{\vec{r}} = \vec{F}_{applied} + \vec{F}_{coriolis} + \vec{F}_{centrifugal}, \tag{54}$$

where the individual forces are,

$$\vec{F}_{applied} = -\nabla V, \tag{55}$$

$$\vec{F}_{coriolis} = 2m\dot{\vec{r}} \times \vec{\Omega} = (2m\Omega\dot{y}, -2m\Omega\dot{x}, 0) \tag{56}$$

$$\vec{F}_{centrifugal} = m\vec{\Omega} \times (\vec{r} \times \vec{\Omega}) = \left(m\Omega^2 x, m\Omega^2 y, 0\right). \tag{57}$$

1.5.2 Motion on a Rotating Earth

Create a primed coordinate system centered at the Earth that is inertial. Fix a point P in the primed coordinate system which will be the location of our rotating laboratory frame. We attach the primed coordinate system basis vectors to the basis vectors at the frame P by identifying them in the following fashion,

$$\hat{x} = \hat{\theta}', \quad \hat{y} = \hat{\phi}', \quad \hat{z} = \hat{r}', \tag{58}$$

meaning that the location of a particle Q measured by P in its rotating frame is given by,

$$\vec{r} = x\hat{\theta}' + y\hat{\phi}' + z\hat{r}', \tag{59}$$

and borrowing the following formulae from the theory of multivariable calculus/curvilinear geometry,

$$\hat{r}' = sin\theta' cos\phi' \hat{x}' + sin\theta' sin\phi' \hat{y}' + cos\theta' \hat{z}', \tag{60}$$

$$\hat{\theta}' = cos\theta' cos\phi' \hat{x}' + cos\theta' sin\phi' \hat{y}' - sin\theta' \hat{z}', \tag{61}$$

$$\hat{\phi}' = -sin\phi' \hat{x}' + cos\phi' \hat{y}', \tag{62}$$

we can replace the polar basis vectors appearing in the expression for $\vec{r}$. Now the location of the particle Q in the reference frame of the inertial observer is given by $\vec{R} + \vec{r}$. This allows us to make comparisons and to write the transformation of coordinates simply as,

$$x' = xcos\theta' cos\phi' - ysin\phi' + (R + z)sin\theta' cos\phi', \tag{63}$$

$$y^{'} = xcos\theta^{'}sin\phi^{'} + ycos\phi^{'} + (R+z)sin\theta^{'}cos\phi^{'}, \tag{64}$$

$$z^{'} = -xsin\theta^{'} + (R+z)cos\theta^{'}. \tag{65}$$

If we allow rotation by a constant angular velocity Ω and call the polar angle the colatitude α we get

$$x^{'} = xcos\alpha cos(\Omega t) - ysin(\Omega t) + (R+z)sin\alpha cos(\Omega t), \tag{66}$$

$$y^{'} = xcos\alpha sin(\Omega t) + ycos(\Omega t) + (R+z)sin\alpha cos(\Omega t), \tag{67}$$

$$z^{'} = -xsin\alpha + (R+z)cos\alpha. \tag{68}$$

Subjecting the above coordinate transformations to the Euler-Lagrange equations gives us a system of equations which can be summarized compactly as,

$$m\ddot{\vec{r}} = m\vec{g} + \vec{F}_{applied} + \vec{F}_{coriolis} + \vec{F}_{centrifugal}, \tag{69}$$

with the corresponding components given as,

$$\vec{g} = (0, 0, -g)\,, \tag{70}$$

$$\vec{F}_{applied} = -\nabla V, \tag{71}$$

$$\vec{F}_{coriolis} = 2m\dot{\vec{r}} \times \vec{\Omega}, \tag{72}$$

$$\vec{F}_{centrifugal} = m\Omega \times \left[(\vec{R} + \vec{r}) \times \vec{\Omega}\right], \tag{73}$$

where $\vec{\Omega}$ is the projection of the angular velocity in the rotating reference frame,

$$\vec{\Omega} = \Omega\hat{z^{'}} = (-\Omega sin\alpha, 0, \Omega cos\alpha)\,. \tag{74}$$

2 Hamiltonian Mechanics

2.1 Hamilton's Equations of Motion

Let us go back to the energy function and consider its variation,

$$dh = dp\dot{q} + pd\dot{q} - dL, \tag{75}$$

$$dh = dp\dot{q} + \cancel{pd\dot{q}} - \frac{\partial L}{\partial q}dq - \cancel{\frac{\partial L}{\partial \dot{q}}d\dot{q}} - \frac{\partial L}{\partial t}dt, \tag{76}$$

$$dh = \dot{q}dp - \frac{\partial L}{\partial q}dq - \frac{\partial L}{\partial t}dt. \tag{77}$$

Keeping this relation handy for a second and considering the energy function as dependent on (q, p, t), we have that

$$dH = \frac{\partial H}{\partial p}dp + \frac{\partial H}{\partial q}dq + \frac{\partial H}{\partial t}dt, \tag{78}$$

and by equating the first two differentials we have that

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}. \tag{79}$$

These are the Hamilton's equations of motion, for the Hamiltonian function,

$$H(p, q, t) = p\dot{q} - L. \tag{80}$$

It should be noted that the Hamilton's Equations of Motion can also be derived from Hamilton's Principle directly, by replacing $L \to H$ and $\dot{q} \to p$ in the action (1) and then doing the usual calculus of variations. Below is the routine.

$$S = \int_t pdq - Hdt, \tag{81}$$

$$\delta S = \int_t \delta dqp + dq\delta p - \left[\frac{\partial H}{\partial q}\delta q + \frac{\partial H}{\partial p}\delta p\right] dt, \tag{82}$$

$$\delta dqp = d\delta qp = d(\delta qp) - \delta qdp, \tag{83}$$

$$\delta S = \int_t d(p\delta q) - \delta qdp + dq\delta p - \left[\frac{\partial H}{\partial q}\delta q + \frac{\partial H}{\partial p}\delta p\right] dt, \tag{84}$$

where the first term disappears after integration. Combining the remaining terms then gives the same system of first order ODES as in (79).

2.2 Conservation Statements

2.2.1 Canonical Momentum

Just like in the case of Lagrangian mechanics, if the Hamiltonian is independent of a coordinate q, then the second of (79) tells us that the canonical momentum is conserved.

2.2.2 Relation Between Time Derivatives

If you paid attention to the last derivation carefully, you would have noticed that the relation

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \tag{85}$$

was absent. It was also proved in the first section on Lagrangian mechanics that

$$\frac{dH}{dt} = -\frac{\partial L}{\partial t}, \tag{86}$$

where h has been replaced by H. We therefore have that,

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \tag{87}$$

Let us see that the first equality also follows.

$$\frac{dH}{dt} = \frac{\partial H}{\partial q}\dot{q} + \frac{\partial H}{\partial p}\dot{p} + \frac{\partial H}{\partial t}, \tag{88}$$

$$\frac{dH}{dt} = \cancel{\frac{\partial H}{\partial q}}\cancel{\frac{\partial H}{\partial p}} - \cancel{\frac{\partial H}{\partial p}}\cancel{\frac{\partial H}{\partial q}} + \frac{\partial H}{\partial t}. \tag{89}$$

2.3 Case Study: Charged Particle in an Electromagnetic Field

The Lagrangian of a charged particle in an electromagnetic field is,

$$L = \frac{1}{2}mv^2 - q\phi + a\vec{A}\cdot\vec{v}, \tag{90}$$

and the corresponding conjugate momentum is,

$$\vec{p} = \frac{\partial L}{\partial \vec{v}} = m\vec{v} + q\vec{A}. \tag{91}$$

The Hamiltonian thus obtained is,

$$H = \vec{p} \cdot \vec{v} - L = \frac{1}{2m}(\vec{p} - q\vec{A}) \cdot (\vec{p} - q\vec{A}) + q\phi. \tag{92}$$

The equations of motion, say, for the x component are,

$$\dot{x} = \frac{\partial H}{\partial p_x} = \frac{1}{m}(p_x - qA_x), \tag{93}$$

$$\dot{p_x} = -\frac{\partial H}{\partial x} = \frac{q}{m}(\vec{p} - q\vec{A}) \cdot \frac{\partial \vec{A}}{\partial x} - q\frac{\partial \phi}{\partial x}. \tag{94}$$

To see that this gives the x component of the Lorentz force law, proceed as follows

$$\frac{d}{dt}(mv_x + qA_x) = \frac{q}{m}(\vec{p} - q\vec{A}) \cdot \frac{\partial \vec{A}}{\partial x} - q\frac{\partial \phi}{\partial x}, \tag{95}$$

$$m\ddot{x} = -q\frac{d}{dt}A_x + \frac{q}{m}(m\vec{v}) \cdot \frac{\partial \vec{A}}{\partial x} - q\frac{\partial \phi}{\partial x}, \tag{96}$$

$$m\ddot{x} = qE_x + q\vec{v} \cdot \frac{\partial \vec{A}}{\partial x} - q\frac{\partial A_x}{\partial x}v_x - q\frac{\partial A_x}{\partial y}v_y - q\frac{\partial A_x}{\partial z}v_z, \tag{97}$$

which conspires in the end to give the familiar

$$m\ddot{x} = q(E_x + (\vec{v} \times \vec{B}))_x. \tag{98}$$

2.4 Poisson Brackets

The Poisson bracket for functions is defined as,

$$[f, g] = \frac{\partial f}{\partial q}\frac{\partial g}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial g}{\partial q}, \tag{99}$$

and it satisfies the property that $[f, g] = -[g, f]$. Below are a collection of results for this mathematical object.

$$[f, q] = -\frac{\partial f}{\partial p}, \quad [f, p] = \frac{\partial f}{\partial q}, \tag{100}$$

$$[q, q] = 0, \quad [q, p] = 1, \quad [p, q] = -1, \quad [p, p] = 0, \tag{101}$$

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + [f, H], \quad \dot{q} = [q, H], \quad \dot{p} = [p, H]. \tag{102}$$

2.5 Liouville's Theorem

We will now see how the density of points in phase space remains a constant of motion. First consider in general the continuity equation,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0, \tag{103}$$

where ρ is the density and $\vec{j} = \rho\vec{v} = \rho(\dot{q}, \dot{p})$ is the current density of phase space. When we unwrap the divergence term we get that,

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho\dot{q})}{\partial q} + \frac{\partial(\rho\dot{p})}{\partial p} = 0, \tag{104}$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial q}(\rho\frac{\partial H}{\partial p}) + \frac{\partial}{\partial p}(-\rho\frac{\partial H}{\partial q}) = 0, \tag{105}$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial \rho}{\partial p}\frac{\partial H}{\partial q} = 0, \tag{106}$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial q}\dot{q} + \frac{\partial \rho}{\partial p}\dot{p} = 0 \implies \frac{d\rho}{dt} = 0. \tag{107}$$

As a consequence of density and total number of points in phase space remaining constant, the total volume occupied by the ensemble of points in phase space also remains constant.

3 Hamilton-Jacobi Theory

Note: The discussion here will limit to a one-dimensional linear pendulum and time-independent transformations. While the extension to more than one dimension in phase space is straightforward, extension to time dependent transformations is performed by considering **infinitesimal canonical transformations**. These remain canonical for infinitely small time intervals, and can be joined together as a sequence to form one big time-dependent canonical transformation.

3.1 Canonical Transformation: Case Study with the Linear Pendulum

Suppose we want to create a transformation from coordinates (q_i, p_i) to (Q_i, P_i) such that the equations of motion remain conserved. Mathematically,

$$\begin{aligned} &Q_i = Q_i(q_i, p_i), \quad P_i = P_i(q_i, p_i), \quad K = K(Q_i, P_i, t) \\ &\dot{Q}_i = \frac{\partial K}{\partial P_i}, \quad \dot{P}_i = -\frac{\partial K}{\partial Q_i}, \end{aligned} \tag{108}$$

and such a transformation is called a **canonical transformation**. In general, the transformation can depend on time, but we will relax this requirement for now. As a first example, let us consider the trivial replacement $Q_i = p_i, P_i = -q_i$,

$$\dot{Q}_i = \dot{p}_i = -\frac{\partial H}{\partial q_i} = \frac{\partial K}{\partial P_i}, \quad \dot{P}_i = -\dot{q}_i = -\frac{\partial H}{\partial p_i} = -\frac{\partial K}{\partial Q_i}, \tag{109}$$

where the equations of motion are conserved in form, but space and momentum have been mixed. As a more physical example, consider the linear pendulum,

$$H = \frac{1}{2}p^2 + \frac{1}{2}w^2q^2, \tag{110}$$

with the equations of motion,

$$\dot{q} = \frac{\partial H}{\partial p} = p, \quad \dot{p} = -\frac{\partial H}{\partial q} = -w^2q. \tag{111}$$

Now consider the transformation,

$$Q = arctan\left(\frac{wq}{p}\right), \quad P = \frac{1}{2w}(p^2 + w^2q^2), \tag{112}$$

with the inverse transformation,

$$q = \sqrt{\frac{2P}{w}}sinQ, \quad p = \sqrt{2wP}cosQ. \tag{113}$$

Let us now show some properties that this transformation fulfills, starting with the Jacobian. Explicitly,

$$|J| = det\begin{vmatrix} \frac{\partial q}{\partial Q} & \frac{\partial q}{\partial P} \\ \frac{\partial p}{\partial Q} & \frac{\partial p}{\partial P} \end{vmatrix} = det\begin{vmatrix} \sqrt{\frac{2P}{w}}cosQ & \frac{sinQ}{\sqrt{2wP}} \\ -\sqrt{2wP}sinQ & \sqrt{\frac{w}{2P}}cosQ \end{vmatrix} = cos^2Q + sin^2Q = 1. \tag{114}$$

What if we calculate the determinant of the Jacobian for the inverse transformation? We expect it to also be equal to unity, but let's see,

$$|J^{-1}| = det\begin{vmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{vmatrix} = det\begin{vmatrix} \frac{pw}{p^2+w^2q^2} & \frac{-wq}{p^2+w^2q^2} \\ wq & \frac{p}{w} \end{vmatrix} = 1. \tag{115}$$

Now we also know that by definition the Poisson Bracket of $[Q, P]_{Q,P} = 1$. What if we calculate $[q, p]_{Q,P}$? We get,

$$[q, p]_{Q,P} = \frac{\partial q}{\partial Q}\frac{\partial p}{\partial P} - \frac{\partial q}{\partial P}\frac{\partial p}{\partial Q} = |J| = 1. \tag{116}$$

By the same token, we also know that by definition the Poisson Bracket of $[q, p]_{q,p} = 1$. What if we calculate $[Q, P]_{q,p}$? We get,

$$[Q, P]_{q,p} = \frac{\partial Q}{\partial q}\frac{\partial P}{\partial p} - \frac{\partial Q}{\partial p}\frac{\partial P}{\partial q} = |J^{-1}| = 1. \tag{117}$$

So the determinant of the Jacobian is unity and the Poisson Brackets remain conserved. So what? Well what we will show now is that the transformation is canonical. First note that the Hamiltonian in the new coordinate system which we call K is,

$$K(Q, P, t) = H(q(Q, P), p(Q, P), t) = wP, \tag{118}$$

and the equations of motion are,

$$\frac{\partial K}{\partial P} = w, \quad -\frac{\partial K}{\partial Q} = 0. \tag{119}$$

Let us see if $\dot{Q} = w$ and $\dot{P} = 0$. For the former we proceed as,

$$tanQ = \frac{wq}{p}, \tag{120}$$

$$(1 + tan^2 Q)\dot{Q} = \frac{w\dot{q}}{p} - \frac{wq}{p^2}\dot{p}, \tag{121}$$

$$(1 + tan^2 Q)\dot{Q} = \frac{w\cancel{p}}{\cancel{p}} - \frac{wq}{p^2}(-w^2 q) = w\left(1 + \frac{w^2 q^2}{p^2}\right) \implies \dot{Q} = w, \tag{122}$$

$$\dot{P} = \frac{1}{2w}(2p\dot{p} + w^2 2q\dot{q}) = 0, \tag{123}$$

confirming that the transformation is canonical. So it seems like a canonical transformation satisfies some nice properties. But why did we bother with such a transformation? The reason is that we have uncovered an ignorable coordinate, and the equations of motion are solved without bothering with a differential equation.

$$P = const, \quad Q(t) = wt + Q_0, \implies q(t) = \sqrt{\frac{2P}{w}} sin(wt + Q_0), \quad p(t) = \sqrt{2wP} cos(wt + Q_0). \tag{124}$$

3.2 General Theory of Canonical Transformations

3.2.1 The Generating Function

Now we ask the question: what must happen to keep the equations of motion looking the same? Recall that the Hamilton equations were derived from the Hamilton's principle, namely that the action must be extremized. So if the equations of motion look covariant, the actions in these two coordinates must be equal up to an exact differential. Mathematically,

$$\delta \int p dq - H dt = \delta \int P dQ - K dt + dF_1, \quad F_1 = F_1(q, Q, t), \tag{125}$$

where F_1 is called the **generating function** of the canonical transformation. The above statement is true because F_1 vanishes at the endpoints of variation and hence does not contribute to the integral. Rearranging the above expression we get that,

$$dF_1 = pdq - PdQ + (K - H)dt, \tag{126}$$

where as from general multivariable chain rule we have that,

$$dF_1 = \frac{\partial F_1}{\partial q}dq + \frac{\partial F_1}{\partial Q}dQ + \frac{\partial F_1}{\partial t}dt. \tag{127}$$

So by comparing terms in the two expansions we arrive at the result,

$$p = \frac{\partial F_1}{\partial q}, \quad P = -\frac{\partial F_1}{\partial Q}, \quad K = H + \frac{\partial F_1}{\partial t}. \tag{128}$$

Notice that if the generating function is time independent, then the old Hamiltonian equals the new Hamiltonian, $K = H$. Now let us see what kinds of generating functions lead to the canonical transformations encountered in the previous section. Consider the generating function $F_1 = qQ$ and Equation (128) in tandem,

$$F_1 = qQ, \implies p = Q, \quad P = -q, \quad K = H, \tag{129}$$

and we can immediately identify this with the first trivial space-momentum mixing studied before. For the linear pendulum case, consider

$$F_1 = \frac{1}{2}wq^2 cotQ, \implies p = \frac{\partial F_1}{\partial q} = \frac{wq}{tanQ}, \quad P = -\frac{\partial F_1}{\partial Q} = \frac{1}{2w}(p^2 + w^2q^2), \quad K = H = wP, \tag{130}$$

which is exactly the transformation considered before. Hence we have found a general condition on a so-called generating function and have constructed the generating functions for our previous two examples.

3.2.2 Alternative Generating Functions

We can replace the independent variables for the generating function. Consider another generating function F_2 defined as,

$$F_2 = F_1 + QP, \tag{131}$$

$$dF_2 = pdq - \cancel{PdQ} + (K-H)dt + \cancel{dQP} + QdP = pdq + QdP + (K-H)dt, \quad (132)$$

such that $F_2 = F_2(q, P, t)$. We have essentially used a Legendre Transformation to replace $Q \to P$. The new conditions are,

$$p = \frac{\partial F_2}{\partial q}, \quad Q = \frac{\partial F_2}{\partial P}, \quad K = H + \frac{\partial F_2}{\partial t}. \quad (133)$$

3.2.3 Direct Conditions

In principle it is not straightforward to construct generating functions, and it is easier to work with the old and new coordinates themselves.

$$\dot{Q} = \frac{\partial Q}{\partial q}\dot{q} + \frac{\partial Q}{\partial p}\dot{p}, \quad (134)$$

$$\dot{Q} = \frac{\partial Q}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial Q}{\partial p}\frac{\partial H}{\partial q} = \frac{\partial K}{\partial P} = \frac{\partial H}{\partial P}, \quad (135)$$

where the second-last equality comes from demanding a canonical transformation, and the last comes from a time-dependent transformation. Expanding further we get that,

$$\frac{\partial Q}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial Q}{\partial p}\frac{\partial H}{\partial q} = \frac{\partial H}{\partial q}\frac{\partial q}{\partial P} + \frac{\partial H}{\partial p}\frac{\partial p}{\partial P}, \quad (136)$$

and by collecting terms we have,

$$0 = \left(\frac{\partial Q}{\partial q} - \frac{\partial p}{\partial P}\right)\frac{\partial H}{\partial p} - \left(\frac{\partial Q}{\partial p} + \frac{\partial q}{\partial P}\right)\frac{\partial H}{\partial q}, \quad (137)$$

which implies that,

$$\frac{\partial Q}{\partial q} = \frac{\partial p}{\partial P}, \quad \frac{\partial Q}{\partial p} = -\frac{\partial q}{\partial P}. \quad (138)$$

Doing the same starting from $\dot{P}$ yields the equations,

$$\frac{\partial q}{\partial Q} = \frac{\partial P}{\partial p}, \quad \frac{\partial p}{\partial Q} = -\frac{\partial P}{\partial q}. \quad (139)$$

As a sanity check, you can see that these statements hold true for the linear pendulum.

3.3 Canonical Invariants

Here we summarize what remains invariant in a canonical transformation.

3.3.1 Jacobian

We had that the determinant of the Jacobian for the linear pendulum transformation was unity. Let us see that it is true for general canonical transformations. First note that,

$$J = \frac{\partial q}{\partial Q}\frac{\partial p}{\partial P} - \frac{\partial q}{\partial P}\frac{\partial p}{\partial Q}, \tag{140}$$

and for the inverse case,

$$J^{-1} = \frac{\partial Q}{\partial q}\frac{\partial P}{\partial p} - \frac{\partial Q}{\partial p}\frac{\partial P}{\partial q}, \tag{141}$$

which under the application of Equations (138) and (139) turns into

$$J^{-1} = \frac{\partial p}{\partial P}\frac{\partial q}{\partial Q} - \frac{\partial q}{\partial P}\frac{\partial p}{\partial Q} = J, \tag{142}$$

giving us that,

$$J = J^{-1}, \implies 1 = J^2 \implies J = 1, \tag{143}$$

that the **volume of phase-space is conserved**.

3.3.2 Poisson Brackets

The Poisson brackets in both these coordinates for arbitrary functions (f, g) are,

$$[f, g]_{q,p} = \frac{\partial f}{\partial q}\frac{\partial g}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial g}{\partial q}, \tag{144}$$

$$[f, g]_{Q,P} = \frac{\partial f}{\partial Q}\frac{\partial g}{\partial P} - \frac{\partial f}{\partial P}\frac{\partial g}{\partial Q}. \tag{145}$$

We can show that the two expressions are identical for a canonical transformation using chain rule and the direct conditions, namely Equations (138) and (139).

3.3.3 Symplectic Condition

We will show that canonical transformations are to a skew-symmetric matrix just as isometries are to a metric. First of all note that the Hamilton equations can be written in the matrix format,

$$\frac{d}{dt}\begin{pmatrix} q \\ p \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} \partial_q \\ \partial_p \end{pmatrix} H \implies \dot{\eta} = S\nabla_\eta H, \tag{146}$$

where S is our skew-symmetric matrix. This also generalizes to higher dimensions in phase space, where S maintains its skew-symmetric structure. Now call ξ the matrix for the new coordinates, just like η was the matrix for the old coordinates. Construct important terms sequentially as follows,

$$\frac{d\xi_i}{dt} = \sum_j \frac{\partial \xi_i}{\partial \eta_j}\frac{\partial \eta_j}{dt} \implies \dot{\xi}_i = \sum_j J_{ij}\dot{\eta}_j, \tag{147}$$

$$(\nabla_\eta)_i = \sum_j \frac{\partial \xi_j}{\partial \eta_i}\partial_{\xi_j} \implies (\nabla_\eta)_i = \sum_j J_{ji}\nabla_{\xi_j}, \tag{148}$$

and the above two expressions in matrix notation reduce to,

$$\dot{\xi} = J\dot{\eta}, \quad \nabla_\eta = J^T\nabla_\xi, \tag{149}$$

and the Hamilton's equations transform as,

$$\dot{\eta} = S\nabla_\eta H \implies J\dot{\eta} = JS\nabla_\eta H \implies \dot{\xi} = JSJ^T\nabla_\xi H. \tag{150}$$

Hence a transformation is canonical iff **the symplectic condition is met**, which is

$$S = JSJ^T. \tag{151}$$

This can be readily verified for the linear pendulum example, since the Jacobian has already been found.

3.4 The Hamilton Jacobi Equation

Our usage of the action functional so far has been limited to its extremization for the equations of motion, and inspection for constraints on canonical transformations. Now we give it the attention it deserves.

3.4.1 Differential Relations for the Action

We have never written the action in differential form. Doing so we get that,

$$\begin{aligned} &dS = pdq - Hdt, \\ &S = S(q,t), \quad p = \frac{\partial S}{\partial q}, \quad -H = \frac{\partial S}{\partial t}. \end{aligned} \tag{152}$$

Let us construct the action for the linear pendulum and verify the above relations. The Lagrangian, equation of motion and general solution are given as,

$$\mathcal{L} = \frac{1}{2}\dot{q}^2 - \frac{1}{2}w^2q^2 \implies \ddot{q} + w^2q = 0 \implies q(t) = q_0 cos(wt) + \frac{\dot{q_0}}{w}sin(wt). \tag{153}$$

The Lagrangian written in terms of the general solution is then,

$$\mathcal{L} = \frac{1}{2}(\dot{q_0}^2 - w^2q_0^2)cos(2wt) - wq_0\dot{q_0}sin(2wt), \tag{154}$$

and evaluating the action gives us,

$$S = \int_0^t \mathcal{L}dt = \frac{(\dot{q_0}^2 - w^2q_0^2)}{2w}sin(wt)cos(wt) + \frac{q_0\dot{q_0}}{2}(cos(2wt) - 1), \tag{155}$$

but before taking partial derivatives we have to replace $\dot{q_0}$. Doing so and taking derivatives confirms the differential relations, namely,

$$\frac{\partial S}{\partial q} = \dot{q} = p, \quad -\frac{\partial S}{\partial t} = H. \tag{156}$$

3.4.2 The Hamilton Jacobi Equation

Using the differential relations for the action, we can write

$$H(p,q,t) + \frac{\partial S}{\partial t} = 0, \tag{157}$$

$$H\left(q, \frac{\partial S}{\partial t}, t\right) + \frac{\partial S}{\partial t} = 0, \tag{158}$$

and is the **Hamilton Jacobi PDE for the action S**. The solution will be of the form.

$$S = S(q, \alpha, t), \tag{159}$$

where α is the constant of integration. The story does not just end here though, and we can actually **use the action as a generating function to completely trivialize our equations of motion!**

$$F_2(q, P, t) = S(q, \alpha, t). \tag{160}$$

Recalling the conditions on the generating function F_2,

$$p = \frac{\partial F_2}{\partial q} = \frac{\partial S}{\partial q}, \quad Q = \frac{\partial F_2}{\partial P} = \frac{\partial S}{\partial \alpha}, \quad K = H + \frac{\partial F_2}{\partial t} = H + \frac{\partial S}{\partial t} = 0. \tag{161}$$

It is the last equation that is so powerful for our purposes. Since the transformation is canonical, we have that

$$\dot{P} = -\frac{\partial K}{\partial Q} = \dot{\alpha} = 0, \tag{162}$$

because we chose $P = \alpha$, it must be constant as expected. But the other coordinate is also constant,

$$\dot{Q} = \frac{\partial K}{\partial P} = 0, \tag{163}$$

because $K = 0$. **Thus both the phase space coordinates (Q,P) remain constant if the action is used as the generating function of a canonical transformation**! This is the power of Hamilton-Jacobi theory, the ability to bypass solving a differential equation. The new position coordinate is now defined to be the constant β. Hence we have moved from (q, p) to (β, α) using the action as a generating function.

3.4.3 Linear Pendulum Case Study Revisited

Now we will sketch out using the Hamilton-Jacobi formalism to solve the linear pendulum example.

$$H = \frac{1}{2}p^2 + \frac{1}{2}w^2q^2 \implies \frac{1}{2}\left(\frac{\partial S}{\partial q}\right)^2 + \frac{1}{2}w^2q^2 + \frac{\partial S}{\partial t} = 0, \tag{164}$$

$$S = W(q) - \alpha t, \tag{165}$$

$$\frac{1}{2}(W')^2 + \frac{1}{2}w^2q^2 - \alpha = 0, \tag{166}$$

$$W' = \sqrt{2\alpha}\sqrt{1 - \frac{w^2 q^2}{2\alpha}}, \tag{167}$$

$$sin(\Phi) = \frac{wq}{\sqrt{2\alpha}} \implies cos(\Phi)d\Phi = \frac{w}{\sqrt{2\alpha}}dq \implies dq = \frac{\sqrt{2\alpha}}{w}cos(\Phi)d\Phi, \tag{168}$$

$$dW = \sqrt{2\alpha}\sqrt{1 - sin^2\Phi}\frac{\sqrt{2\alpha}}{w}cos(\Phi)d\Phi, \tag{169}$$

$$dW = \frac{2\alpha}{w}cos^2\Phi d\Phi, \tag{170}$$

$$W = \int \frac{\alpha}{w}\left(1 + cos(2\Phi)\right)d\Phi = \frac{\alpha}{w}\left(\Phi + sin\Phi cos\Phi\right), \tag{171}$$

$$S = \frac{\alpha}{w}\left(\Phi + sin\Phi cos\Phi\right) - \alpha t, \tag{172}$$

$$\beta = \frac{\Phi}{w} - t \implies \Phi = w(\beta + t) \implies q(t, \alpha, \beta) = \frac{\sqrt{2\alpha}}{w}sin(w(t + \beta)). \tag{173}$$

3.4.4 Action-Angle Variables

Now we will wrap up Hamilton-Jacobi theory by studying periodic motion. The Hamiltonian is typically conserved in such systems so it is a constant, independent of time. Start by inverting the Hamiltonian to find p as,

$$H(q, p) = \alpha \implies p = p(q, \alpha), \tag{174}$$

and define the **action angle variable** J as,

$$J = \int_T pdq \implies \alpha = H(J), \tag{175}$$

meaning that the Hamiltonian α is a function of J, which itself is also a constant. Now we insert this action angle variable in the slot for the constant in the generating function $F_2(q, J)$. Notice that this is a time-independent generating function, so the Hamiltonian $K = H$. Doing so we have that,

$$p = \frac{\partial F_2}{\partial q}, \quad \theta = \frac{\partial F_2}{\partial J}, \tag{176}$$

where θ is the new spatial coordinate. Now by virtue of a canonical transformation, we have that

$$\dot{\theta} = \frac{\partial H}{\partial J} = \nu, \quad \dot{J} = -\frac{\partial H}{\partial q} = 0, \tag{177}$$

where the last expression is expected. The first expression defines another constant ν. The solution is then,

$$\theta = \nu t + \beta, \quad J = constant. \tag{178}$$

To appreciate the naming convention, look at the change of w over one period,

$$\Delta\theta = \int_T \frac{\partial w}{\partial q} dq, \tag{179}$$

$$\Delta\theta = \int_T \frac{\partial^2 F_2}{\partial q \partial J} = \frac{\partial}{\partial J} \int_T p dq = 1, \tag{180}$$

$$1 = \Delta\theta = \nu T \implies \nu = \frac{1}{T}, \tag{181}$$

meaning that θ changes by 1 over a period with frequency ν. Consider again the linear pendulum, which is periodic.

$$H = \alpha = \frac{p^2}{2} + \frac{w^2 q^2}{2} \implies p = \sqrt{2\alpha - w^2 q^2} \implies J = \frac{2\pi\alpha}{w}, \tag{182}$$

$$H(J) = \alpha = \frac{wJ}{2\pi}, \quad \nu = \frac{\partial H}{\partial J} = \frac{w}{2\pi}, \tag{183}$$

$$\dot{\theta} = \nu = \frac{w}{2\pi}, \tag{184}$$

which is just a statement that w is the angular frequency.